EXP NO:16 Experiment to understand EDA-Quantitative and Qualitative analysis.

AIM:
To understand the importance of EDA - Quantitative and Qualitative analysis.

PROGRAM:

python

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# Read dataset
data = pd.read_csv(
    'https://raw.githubusercontent.com/mwaskom/seaborn-data/master/tips.csv'
)
data

# Display first 5 rows
data.head()

# Information about the dataset
data.info()

# Check missing values
data.isnull().sum()

# ----- Quantitative analysis -----

# Statistical description of numerical columns
data.describe()

# Mean, median and standard deviation of tip
print("Mean:", data['tip'].mean())
print("Median:", data['tip'].median())
print("Standard Deviation:", data['tip'].std())

# Correlation between numerical columns
data[['total_bill', 'tip', 'size']].corr()

# Heatmap of correlation
sns.heatmap(data[['total_bill', 'tip', 'size']].corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()

# Histogram of total bill
sns.histplot(data['total_bill'], kde=True)
plt.title("Distribution of Total Bill")
plt.show()

# Boxplot of tip
sns.boxplot(x=data['tip'])
plt.title("Boxplot of Tip")
plt.show()

# ----- Qualitative analysis -----

# Unique values in categorical columns
print(data['day'].unique())
print(data['sex'].unique())
print(data['smoker'].unique())
print(data['time'].unique())

# Count of each category
data['day'].value_counts()
data['sex'].value_counts()

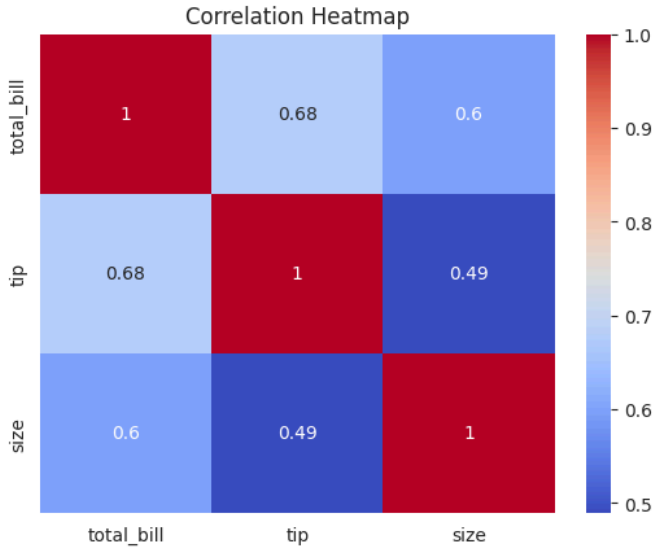
# Countplot of day
sns.countplot(x='day', data=data)
plt.title("Number of Records per Day")
plt.show()

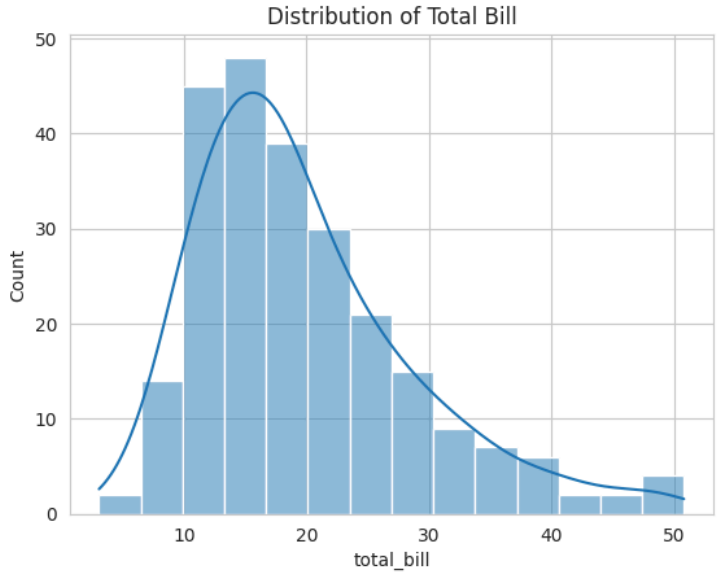
# Countplot of day by time
sns.countplot(x='day', hue='time', data=data)
plt.title("Day vs Time")
plt.show()

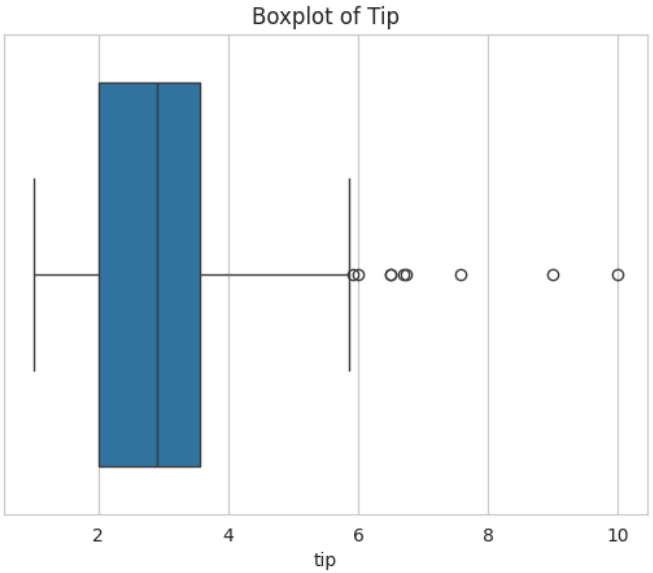
# Average tip for each day
data.groupby('day')['tip'].mean()

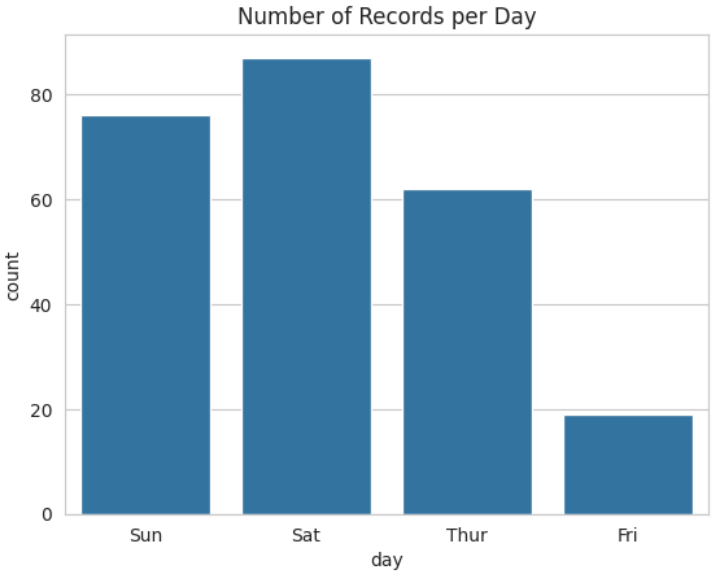
# Boxplot of tip by day
sns.boxplot(x='day', y='tip', data=data)
plt.title("Tip by Day")
plt.show()
```

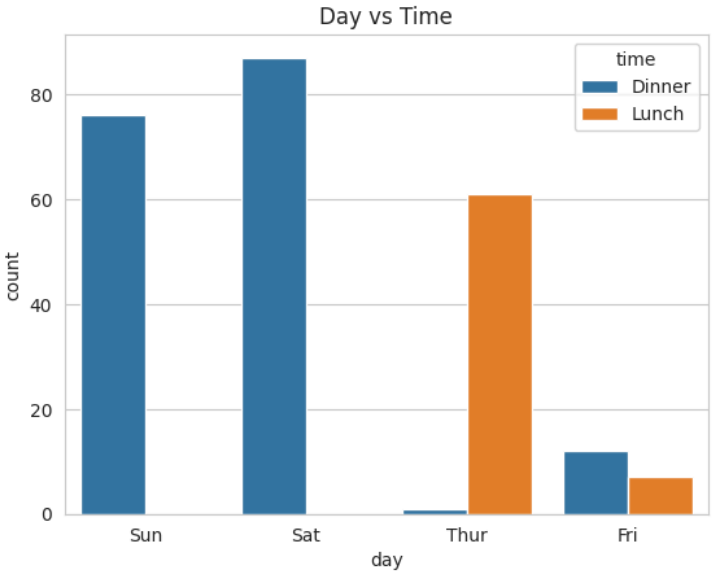
OUTPUT:

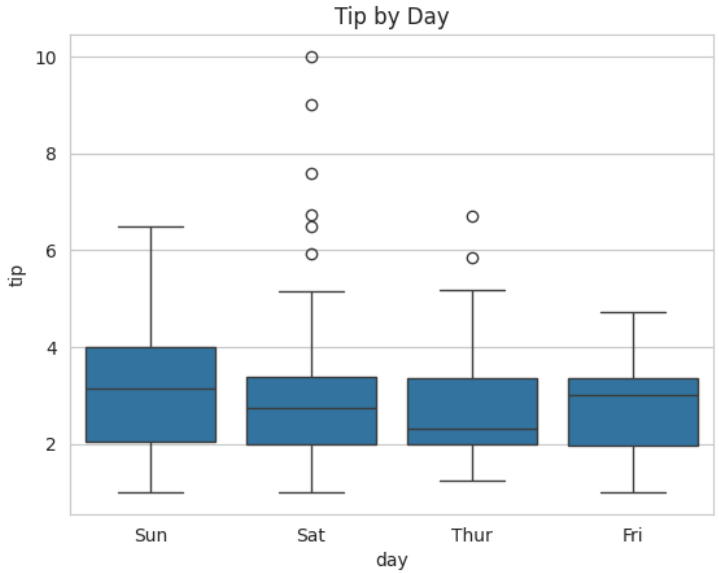

Correlation Heatmap
total_bill
tip
size
1
0.68
0.6
0.68
1
0.49
0.6
0.49
1
total_bill
tip
size
1.0
0.9
0.8
0.7
0.6
0.5


Distribution of Total Bill
Count
total_bill
0
10
20
30
40
50
10
20
30
40
50


Boxplot of Tip
2
4
6
8
10
tip


Number of Records per Day
0
20
40
60
80
count
Sun
Sat
Thur
Fri
day





RESULT:

Thus the program to understand Exploratory Data Analysis using quantitative and qualitative analysis was executed successfully.